

pip

pip is the package manager for Python. Using pip we can install python packages.

How to Install pip

pip is distributed with Python which means that when you download Python from <https://www.python.org/> , you automatically get pip installed on your computer.

Check if pip is already Installed

```
pip --version
```

Update pip

```
python -m pip install --upgrade pip
```

```
py -m pip install --upgrade pip
```

How to Get Help

pip help – It shows list of pip commands and their functions.

How to install Packages

pip install camelcase

How to Use Packages

import camelcase

How to Uninstall Packages

pip uninstall camelcase

numpy

In Python, Numpy is a package which contains classes, functions, variables, large library of mathematical functions etc to work with scientific calculation.

Numpy can be used to create n dimensional arrays where n is any integer. We can create 1 dimensional array, 2 dimensional array, 3 dimensional array and so on.

Numpy's array class is called ndarray. It is also known by alias name array. There is another class array in python which is different from numpy's array class.

Import numpy

There are two ways to import numpy:-

- `import numpy` – This will import the entire numpy module.
- `from numpy import *` - This will import all class, objects, variable etc from numpy package. Here * means All.

One Dimensional Array

Single Row Multiple Columns

Ex:- [101, 102, 103, 104, 105]

Ways of Creating Array in numpy

- array () Function
- linspace () Function
- logspace () Function
- arange () Function
- zeros () Function
- ones () Function

array () Function

array () Function of numpy module is used to create an array.

Syntax:-

```
numpy.array(object, dtype=None, copy=True, order='K', subok=False, ndmin=0)
```

<https://www.numpy.org/devdocs/reference/generated/numpy.array.html>

Creating 1D Array using array () Function

Syntax:-

```
import numpy
```

```
array_name = numpy.array([elements])
```

array Function



Ex:-

```
import numpy
```

```
stu_roll = numpy.array([101, 102, 103, 104, 105])
```

```
stu_roll = numpy.array([101, 102, 103, 104, 105], int)
```

```
a = numpy.array([10.1, 5.2, 4.23, 10.4, 1.5], float)
```

```
a = numpy.array([10.1, 5.2, 4.23, 10.4, 1.5])
```

```
ar = numpy.array(['a', 'b', 'c', 'd'])
```

```
name = numpy.array(['Rahul', 'Sonam', 'Raj'], dtype=str)
```

```
name = numpy.array(['Rahul', 'Sonam', 'Raj'])
```

Creating 1D Array using array () Function

Syntax:-

```
from numpy import *  
array_name = array([elements])
```

Ex:-

```
from numpy import *  
stu_roll = array([101, 102, 103, 104, 105])
```

index

An index represents the position number of an array's element.

```
from numpy import *
```

```
stu_roll = array([101, 102, 103, 104, 105])
```

Python interpreter allocates 5 blocks of memory and stores the elements

Index always starts with 0



101	102	103	104	105
0	1	2	3	4

101	102	103	104	105
<code>stu_roll[0]</code>	<code>stu_roll[1]</code>	<code>stu_roll[2]</code>	<code>stu_roll[3]</code>	<code>stu_roll[4]</code>

Accessing One-D Array Elements

```
from numpy import *
```

```
stu_roll = array([101, 102, 103, 104, 105])
```

```
print(stu_roll[0])
```

```
print(stu_roll[1])
```

```
print(stu_roll[2])
```

```
print(stu_roll[3])
```

```
print(stu_roll[4])
```

101	102	103	104	105
<i>stu_roll[0]</i>	<i>stu_roll[1]</i>	<i>stu_roll[2]</i>	<i>stu_roll[3]</i>	<i>stu_roll[4]</i>

Modifying 1D Array Elements

```
from numpy import *
```

```
stu_roll = array([101, 102, 103, 104, 105])
```

```
stu_roll[1] = 110
```

```
print(stu_roll[1])
```

101	102	103	104	105
stu_roll[0]	stu_roll[1]	stu_roll[2]	stu_roll[3]	stu_roll[4]

Accessing array using for loop

```
from numpy import *  
stu_roll = array([101, 102, 103, 104, 105])
```

Without index

```
for element in stu_roll:  
    print(element)
```

With index

```
n = len(stu_roll)  
for i in range(n):  
    print(stu_roll[i])
```

Accessing array using while loop

```
from numpy import *
```

```
stu_roll = array([101, 102, 103, 104, 105])
```

```
n = len(stu_roll)
```

```
i = 0
```

```
while i < n :
```

```
    print(stu_roll[i])
```

```
    i+=1
```